

APPROVED



MSc Cyber Security
School of Computing, Engineering and Technology

Version	5.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	
No Course Award Assigned	
Code	0558
SCQF Level	Level 11
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September as of (September 2026)

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100365	computer networks	computer networks	100%
Total			100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

What the Course Involves	The general format of the first two semesters for full-time students comprises a mix of lectures and lab sessions. Students will have access to all materials on the University's VLE. In semester 3, students are engaged on a project module, and have a structured timetable as well as meetings with their project supervisor to discuss progress and offer support. Part-time students attend the same programmed sessions as the full-time students but over an extended duration corresponding to two 15-credit modules per semester. Full Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January)
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	Part Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (taken in September), Semester 4 (taken in January), Semester 5 (project, taken in May and spanning 2 semesters) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 4 (taken in January), Semester 3 (taken in September), Semester 5 (project, taken in January and spanning 2 semesters)
Course Collaborations	
No collaborations listed	
Introduction to the Course	The MSc/PgD/PgC Cyber Security is a specialist Masters course, designed to equip Honours Computing graduates with expert knowledge of information and network security technology and to provide the foundation research skills required to become an independent researcher. It provides the technical skills and training necessary to develop secure information systems and manage complex networks to meet the challenge required in merging technologies. The technical content of the course is aimed at relevant industrial techniques and theoretical underpinning in specialized areas. The MSc project is of particular value to employers as evidence of the student's ability to produce high-quality work of an advanced technical standard to deadline and to a high standard of presentation. The course includes a variety of practically-focused security subjects, reflecting the strong demand from industry for graduates with hands-on skills in this area. These subjects cover the full spectrum of the security programme including an appreciation of the security threat landscape and the management of risk, the setting and management of security policies and controls across the enterprise IT infrastructure (systems and networks), the testing of security readiness and the handling of security incidents. Students will be working in a dedicated cyber security lab. At all stages of study, the course content is complemented by practical work intended to reinforce the learning process but also to enhance the vocational relevance of the programme of study. The syllabus content also incorporates modules intended to enhance the technical and professional skills required to manage and execute state-of-the-art computer systems.
	The aims of the course are to: • provide specialist, advanced technical skills in the areas of cyber security; • produce specialist practitioners in the specification, design and implementation of secure information systems and networks; • provide students with the knowledge and skills required to manage computer facilities with a view to preventing, detecting and recovering from security breaches; • enable students to access, critically appraise and disseminate research results; and • provide students with a sound basis for further research and / or professional development. Aims of Each Stage of the Course The specific aims for each stage of the course are: Postgraduate Certificate To produce practitioners who will be able to: • demonstrate knowledge and understanding of current information system security concepts and technologies; • test and evaluate the security of an IT infrastructure.

Educational Aims of the Course	Postgraduate Diploma To produce practitioners who, in addition to the aims above, will be able to: • apply the methods and techniques used in designing and managing secure database systems and web applications; • design and implement machine learning solutions for defensive and offensive security; • critically appraise the usability criteria of security and privacy; • demonstrate knowledge and understanding of the key role of human factors in cyber security; • investigate, and use appropriate methods to contain and recover from, a security incident; • develop practical project planning and research skills; and foster an awareness of professional, ethical and legal implications of IT; • effectively communicate, in speech and writing, the ideas inherent in advanced technical systems. Master of Science In addition to the above, MSc graduates will be able to: • apply their expertise in their chosen area of specialised study; • plan, organise and execute a substantial investigative project within the topics of the course; • critically appraise different approaches and technical solutions; • critically appraise / test and document the project outcomes.
Awarding Institution	Robert Gordon University
Course Duration	Full-Time Students Full-time 12 months with a September intake Full-time 16 months with a January intake Part-Time Students Part-time 28 months with a September intake Part-time 32 months with a January intake Depending on their mode of study, students can choose to undertake either a short-term (3 months) or long-term (12 months) placement. All placements require approval from the School prior to commencement. If a student goes on placement, the duration of the course will be extended by the duration of the placement.
Distinctive Features of the Course	The course offers specialist technical skills in the area of cyber security. The course content is professionally relevant and is related to key research in the School. The course includes both a long placement and a short placement that offer students the opportunity to apply and further develop their technical skills, and build teamwork skills in a realistic business environment.
	Students may elect to undertake a short placement or a year-long placement.

Placement/Professional Experience	The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who the student can contact about any academic or non-academic problems during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 3 months of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor. As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.
Opportunities for Further Study	The course has a strong grounding in research techniques and enables students to advance into research and development at doctoral level within this institution or other UK higher education institutions.
	SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the National Cyber Security Centre (NCSC) Cyber Security Body of Knowledge (CyBOK) and British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind, and embedded in the core structure are modules and topics which provide the student with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society

External and Internal Indicators of Quality and Standards	in order to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's programmes. A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Programme Management Teams, Student Experience questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the University quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University quality assurance committee structures, such as the University's Student Experience, and Academic Regulations Sub-Committees. Finally, it should be mentioned that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of computing science education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computer science education, as well as more general work on student transitions. The School also participates in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Applicants normally hold a 2:2 undergraduate honours degree or its equivalent in a subject area with substantial Computing content. Applicants whose first language is not English should have an IELTS of 6.0 overall, with no component lower than 5.5, or an equivalent.
Alternative Qualifications for Entry	Applicants with a degree in an alternative subject area or lower degree classification and/or with relevant work experience will be considered on a case by case basis.
Interview Procedures	Generally, an interview is not required for admission to the course.

Course-Specific Regulations	The MSc Project module cannot be condoned as it is key to demonstrating essential skills in the course such as the ability to conduct a significant piece of independent work. Furthermore, BCS guidelines state that the 60 credit project should not be compensated. The placement modules cannot be used to compensate for failed modules. In order to meet the requirements of the course's full certification by the National Cyber Security Centre (NCSC), where an assessment uses percentage marks as the basis of obtaining an overall grade, the pass mark is required to be at least 50%. All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK.

	RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	The extent of the course syllabus demands a relatively rapid pace of delivery. Consequently, student-centred learning strategies play an important role in most subjects and throughout all stages, reinforcing and supplementing more traditional methods of subject delivery. Directed reading, independent progress through computer-based exercises, independent reading, analysis of case studies and directed searches of on-line information resources are all commonplace. The preferred strategy for any course unit depends on the nature and extent of the syllabus content, on its subject-specific learning outcomes and its contribution to the overall course aims. In the sections that follow, a more detailed description of the learning experience is given, which distinguishes between: • Core technological modules; • Integrative course modules. N.B. Teaching and learning are reviewed through an annual appraisal mechanism and strategies are liable to change for the sake of continuing improvement and to reflect best practice. Technical report writing skills are exercised in all modules. The course addresses project planning skills that are vocationally relevant and that also directly underpin the planning and management of the MSc Project. Student feedback in this course takes the form of critical self-appraisal, peer appraisal, and commentary from academic staff. Through directed reading and by listening to invited external speakers, students are also encouraged to develop a broader perspective of Information Technology and to sharpen their awareness of the need for professional standards and ethics in the workplace. <u>Core Technological Units</u> These modules impart specialist technical knowledge through the medium of lectures and tutorials, supplemented as necessary by student-centred material and tasks. All core modules have a supporting practical element, which is delivered through laboratory work. In practical sessions students are expected to progress through a series of graded exercises that are designed to foster analytical skills and to illustrate the synthesis of fundamental knowledge and problem solving techniques in a structured problem domain. Students will be working in dedicated networking and cyber security labs. These modules will provide key concepts in the main security domains and will equip the students with the necessary knowledge and skills in securing information systems and networks as well as detect and respond to security incidents. <u>Integrative Course Unit</u>

	The MSc Project module serves the primary purpose of integrating technological strands, which are developed in the preceding PgDip stage, and do so in the context of a substantial cyber security project. The project typically involves all stages of a substantial research project including (i) an investigation of a technical problem and its domain, (ii) examining relevant methods and tools, (iii) gaining an understanding of the problem and its domain and developing a project specification, and (iv) the development and evaluation of a solution to a problem. The problem is original to the student and its solution therefore requires the innovative application of knowledge and techniques either studied in the previous PgD stage or acquired through independent research of recent and relevant literature. The MSc Project provides a vehicle for integrating specialist knowledge with analytic, problem solving, managerial and communication skills. All of these are exercised and evidenced through the execution and outcomes of the project. <u><i>Research Skills</i></u> The course framework underpins the MSc Project in a number of ways. The PgC/PgD taught modules provide support in specific subject areas. Students also require competence in project management, research skills and the professional, legal, ethical and social issues in Information Technology. These skills are intended to assist the student in the investigation of a problem domain and the appraisal of appropriate research methods and techniques that may be relevant to the achievement of the project goals. Beyond the confines of the MSc Project, these same research skills are useful for continuing professional development and act as a safeguard against technological obsolescence. All MSc students are also invited and encouraged to participate in the School's weekly seminar where industrialists and researchers present their work.
Teaching and Learning Strategy	The course curriculum is designed to enable the student to acquire and develop their subject knowledge and understanding, and to gain practical and key transferable skills at Masters level. Specifically to practice and apply their knowledge skills and understanding, critically apply cognitive skills, develop and practice a wide range of communications skills, and exercise autonomy and accountability when working with others, consistent with the corresponding Scottish Credit and Qualifications Framework level 11 guidelines. As the students progress through the course they are encouraged to expand their understanding and critical appreciation of advanced computing, and develop skills in programming through techniques and tools with the application of this knowledge. The overall philosophy and strategy for the course is to provide a sound knowledge and understanding of advanced computing in the broader context. The vocationally orientated nature of the course requires the student to achieve general and specific skills, which are addressed through a combination of teaching and learning methods, including lectures, lab sessions, tutorials, seminars, student-centred learning and private study (directed reading of textbooks and online material) and structured laboratory sessions. The broad nature of the programme including common and specialist elements necessitates the use of a broad range of teaching techniques. These include lectures, seminars, tutorials, computer laboratories, open and distance learning and group work on a large project. Directed guest lectures are used to encourage and emphasise professional skills. Course induction and directed study promote development and enhancement of skills in subject context throughout course. A handbook on basic study skills is provided to all students. Learning skills are developed through tutorials, directed reading and projects. Tutorials, case studies and graded problem assignments e.g. in laboratory sessions are employed to provide practical and theoretical underpinning of subject matter provided by lectures. Students meet regularly with project supervisors during MSc project modules.

Assessment	A range of assessment methods are used, with particular emphasis placed on coursework assignments, including design exercises (ranging from design studies to fully implemented solutions), comparative studies, and investigative reports. Students are told in advance what assessment methods will be used for each module and will have access to examples of past assessments. Where the format of assessment varies significantly from one year to the next, examples of assessment will be created in order to ensure students understand what the assessment entails. All assessments include a marking/grading scheme which describes the criteria used for the award of marks/grades. Structured project documents will specify the management of the project conduct, quality guidelines and project deliverables. It will also contain information on the use of web resources to present interim MSc Project progress updates and guidelines on the live presentation and demonstration of MSc Project outcomes. The use of IT tools will be assessed in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework). Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	Offering 100% wireless network coverage, the School has a total of 335 desktop workstations in 9 IT labs. Apart from a Mac lab of 30 machines, all workstations are currently running Windows. All the desktops are replaced on a 3 year basis. The IT facilities include a Green Screen Suite used in multimedia video production, a dedicated CISCO networking lab and a Cyber Security Lab. A core suite of applications are installed on all workstations. The School is a member of Microsoft Dreamspark Premium and the VMware academy which allows the students access to commercial software, including installation on their own machines. In addition, the School's virtual learning environment (CampusMoodle) offers 24/7 access to all teaching materials. The School has its own server racks operating within the two data centres used by the university which are home to Windows HPC servers, Apple servers, ESX servers and accompanying Storage Area Network, and a Boston 8 node render farm. These servers (and all the computer labs) are maintained by the School's dedicated team of systems administrators. Access to the School's labs is by electronic key fob at any time during University opening hours. Students have access to the University's wireless network. Students also have remote access through Citrix Desktop.

	All students are given access to the Student handbook, which contains a wide variety of information about the School and University. Students will be supported through the School's integration of the institutional Team Around a Student (TAAS) approach, ensuring that every student has a point of contact who can provide advice, triage queries, and connect them with the most appropriate support and resources where issues may be affecting their studies or wider student experience. The weekly induction sessions during the first semester of study cover a wide variety of topics such as library facilities, extenuating circumstances, exam preparation, careers services, etc. All students are assigned a supervisor for their project work. In addition, they receive extensive guidelines on how to undertake a project.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	The School is committed to the highest standards of quality and best pedagogy and research within an environment which values all associated stakeholders. Formal feedback is gathered from Student Questionnaires which are issued on completion of the PgDip stage of the course. It offers the students the opportunity to comment in detail on all aspects of the course as well as the general environment, support services and facilities. Responses to, and actions resulting from, these comments are addressed fully in an Annual Course /Programme Appraisal and widely discussed by the appropriate Programme Management Team. The Course Team also runs a series of informal focus groups in order to gather both positive and negative comments on the course. Where problems are highlighted, these are discussed fully and potential solutions are considered. Less formally, the students have an opportunity to discuss with their Course Leader any course or management issues.
	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements.

University Student Feedback Mechanisms

This approach is underpinned by the *Student Partnership Agreement* which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: [Student Representation & Partnership | RGU](#) .

There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience.

Mechanisms through which students are engaged and supported include:

- participation in student evaluation questionnaires and where relevant the *National Student Survey (NSS)* ;
- staff/student liaison arrangements at course/programme level;
- support for Student Representatives;
- student representation on Institution-Led Subject Review Panels and Validations;
- student representation on key University committees.

Course-Specific Student Feedback Mechanisms

The University uses a variety of mechanisms to obtain feedback from students about their learning experience. The University has a system of student representation, via the appointment of Student Representatives for each course, who in turn represent the interests of their colleagues at Staff/Student Liaison meetings held each semester with staff representatives from the School concerned. Student Representatives are also invited to attend a meeting with the Dean of School during the first and second semester to discuss a selection of topics and to air any concerns they have. The notes of all these meetings and responses to any issues raised are subsequently made available.

A single Staff/Student Liaison Committee exists for all postgraduate courses in the School, which meets regularly. The Committee comprises the Dean of School, Course Leaders, teaching staff and class representatives from each course intake and meets at least once per semester to exchange information and discuss issues.

Awards**Stage 1**

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Postgraduate Certificate - PgCert	Cyber Security	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	Cyber Security	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	Cyber Security	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate knowledge and understanding of information security management, ethical hacking, systems and network management and security, database and web security, and digital forensics;	SC1	SCQF 11
LLO2	Demonstrate knowledge and understanding of tools and techniques which aid the specification, design and implementation of secure information systems and networks;	SC1	SCQF 11
LLO3	Demonstrate knowledge and understanding of characteristics and capabilities of network management tools;	SC1	SCQF 11
LLO4	Demonstrate knowledge and understanding of: the evaluation of the security implications of computer networks and develop a security policy to protect networks, systems and data;	SC1	SCQF 11
LLO5	Demonstrate knowledge and understanding of suitable file and data encryption mechanisms to prevent eavesdropping and protect privacy;	SC1	SCQF 11
LLO6	Demonstrate knowledge and understanding of the definition and implementation of counter measures to combat against unauthorised network access;	SC1	SCQF 11
LLO7	Demonstrate knowledge and understanding of the role of human behaviour in security;	SC1	SCQF 11
LLO8	Demonstrate knowledge and understanding of the machine learning techniques used in cyber security;	SC1	SCQF 11
LLO9	Demonstrate knowledge and understanding of: the selection and establishment of a project management framework and lay down effective procedures for monitoring progress of any large scale research / development project;	SC1	SCQF 11
LLO10	Demonstrate knowledge and understanding of the availability of information resources and how to access these to obtain state-of-the-art knowledge of current information systems technology;	SC1	SCQF 11
LLO11	Demonstrate knowledge and understanding of: research methodology appropriate to this level of work;	SC1	SCQF 11
LLO12	Demonstrate knowledge and understanding of research methods appropriate specifically to their Masters' independent study project, together with detailed knowledge of the particular area or sub-area in which the project is carried out;	SC1	SCQF 11
LLO13	Demonstrate knowledge and understanding of the theoretical and empirical limits and boundaries of the area(s) in question, and of the range of methods of study and types of judgements employed by advanced practitioners;	SC1	SCQF 11
LLO14	Demonstrate knowledge and understanding of appropriate research methodology and detailed understanding of the particular area or sub-area in which the Masters' project is carried out.	SC1	SCQF 11
LLO15	Demonstrate the skills required to conduct a security test to systems, applications and networks;	SC2	SCQF 11
LLO16	Demonstrate the skills required to investigate a security incident and conduct its related forensics work.	SC2	SCQF 11
LLO17	Demonstrate the skills required to implement and manage secure systems, applications and networks;	SC2	SCQF 11
LLO18	Demonstrate the skills required to apply statistical methods to assess and evaluate research or experimental test results.	SC2	SCQF 11
LLO19	Demonstrate the skills required to make effective use of available IT tools and packages to undertake the practical work of the PgC/PgD modules, to produce high quality technical documentation to accompany coursework submissions and to access online learning resources.	SC2	SCQF 11
LLO20	Demonstrate the skills required to make effective use of available IT tools and packages to enhance the production and presentation of the Masters stage project report and oral defense.	SC2	SCQF 11
LLO21	Demonstrate the skills required to produce structured and coherent written reports to document a problem investigation and research	SC2	SCQF 11

LLO22	Demonstrate the skills required to convey strategy/design decisions and record research/test strategies and results throughout the various stages of a major research project.	SC2	SCQF 11
LLO23	Demonstrate the skills required to work effectively as part of a team, sharing information with other team members and producing appropriate documentation.	SC2	SCQF 11
LLO24	Demonstrate the skills required to produce a clear and coherent thesis which documents the process and end result of a major research /information systems development exercise.	SC2	SCQF 11
LLO25	Demonstrate the skills required to deliver a coherent and lucid thesis defense before a panel.	SC2	SCQF 11
LLO26	Work autonomously or with minimal guidance, where appropriate, and manage own learning using the full range of resources and study techniques appropriate to the specialist area(s) covered by the course.	SC3	SCQF 11
LLO27	Demonstrate the various skills required to work effectively with a research supervisor and with any other support staff;	SC3	SCQF 11
LLO28	Identify and analyse problem requirements.	SC3	SCQF 11
LLO29	Effectively implement and manage security controls	SC3	SCQF 11
LLO30	Demonstrate the ability to propose suitable solutions for the design and implementation of secure systems, applications and networks.	SC3	SCQF 11
LLO31	Demonstrate the ability to evaluate the efficacy of proposed solutions.	SC3	SCQF 11
LLO32	Apply critical judgment in the selection and application of analysis/test strategies for validating research outcomes/testing software and compliance to initial requirements.	SC3	SCQF 11
LLO33	Demonstrate the ability to formulate effective solutions to realistic problems related to the focus of the course.	SC3	SCQF 11
LLO34	Demonstrate the ability to collaborate with other team members in order to deliver a joint solution to a problem.	SC3	SCQF 11
LLO35	Demonstrate the ability to select and conduct the Masters project applying an analytic, rigorous and critical approach to identifying project goals, proposing and evaluating different research/solution strategies and generating a useful and effective output.	SC3	SCQF 11
LLO36	Demonstrate the ability to develop and evaluate effective solutions to computer and network security problems.	SC4	SCQF 11
LLO37	Demonstrate the ability to apply advanced skills to testing the security of systems, applications and networks.	SC4	SCQF 11
LLO38	Demonstrate the ability to investigate and critically appraise specialist areas of research.	SC4	SCQF 11
LLO39	Demonstrate the ability to integrate critical reflection into work roles and responsibilities.	SC4	SCQF 11
LLO40	Demonstrate the ability to consider and assess ethical and legal issues with respect to own work.	SC5	SCQF 11
LLO41	Demonstrate the ability to adopt an analytical approach to tasks to identify problems, solutions and evaluation.	SC4	SCQF 11
LLO42	Demonstrate the ability to work as an effective team member.	SC5	SCQF 11
LLO43	Demonstrate the ability to express complex technical information verbally and through written and visual work;	SC4	SCQF 11
LLO44	Demonstrate the ability to apply a range of learning techniques	SC4	SCQF 11
LLO45	Show a willingness to learn from others and problem solve.	SC5	SCQF 11
LLO46	Demonstrate the ability to work independently, self manage time and workload scheduling flexibility, critically analyse, and objective evaluation.	SC5	SCQF 11
LLO47	Demonstrate the ability to work as part of a team, sharing work and contributing effectively to the completion of the task at hand.	SC5	SCQF 11
LLO48	Demonstrate the ability to challenge, test, analyse and evaluate programming issues and work toward a realistic and workable solution.	SC4	SCQF 11

LLO49	Demonstrate the ability to thoroughly evaluate work in an objective manner.	SC4	SCQF 11
LLO50	Demonstrate the ability to conduct an ethical and legal assessment of proposed work.	SC5	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A - Non-Placement Route B - Short Placement Route C - Long Placement
Part Time On Campus	Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Stage 1 / Semester 1**Full Time On Campus**

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Part Time On Campus

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CMM542	Human Factors in Security		Christopher McDermott (cdm7692)	Yes	2.0	15	Level 11
Part Time On Campus	CMM517	Information Security Management		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11
Full Time On Campus	CMM517	Information Security Management		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11
Full Time On Campus	CMM541	Machine Learning for Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 11
Part Time On Campus	CMM518	Security Testing		Muhammad Shadi Hajar	Yes	1.0	15	Level 11
Full								

Time On Campus	CMM518	Security Testing		Muhammad Shadi Hajar	Yes	1.0	15	Level 11
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Stage 1 / Semester 2**Full Time On Campus**

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Part Time On Campus

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CMM523	Database and Web Security		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11
Full Time On Campus	CMM523	Database and Web Security		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11
Full Time On Campus	CMM519	Incident Management and Forensics		Ian Harris (ish7850)	Yes	2.0	15	Level 11
Part Time On Campus	CMM528	Network Security		Christopher McDermott (cdm7692)	Yes	1.0	15	Level 11
Full Time On Campus	CMM528	Network Security		Christopher McDermott (cdm7692)	Yes	1.0	15	Level 11
Full Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11

Stage 1 / Semester 3**Full Time On Campus**

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Part Time On Campus

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Part Time On Campus	CMM542	Human Factors in Security		Christopher McDermott (cdm7692)	Yes	2.0	15	Level 11
Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Part Time On Campus	CMM541	Machine Learning for Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 11
Elective								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Stage 1 / Semester 4**Part Time On Campus**

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Part								

Time On Campus	CMM519	Incident Management and Forensics		Ian Harris (ish7850)	Yes	2.0	15	Level 11
Part Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11

Stage 1 / Semester 5**Part Time On Campus**

Route A - Non-Placement Route B - Short Placement Route C - Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CEM100	MSc Project	Routes A, B, C; undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Part Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Full Time On Campus	CMM542	Human Factors in Security		Christopher McDermott (cdm7692)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time On Campus	CMM517	Information Security Management		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM517	Information Security Management		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM541	Machine Learning for Cyber Security		Harsha Kalutarage	Yes	1.0	15	Level 11

			Campus				(hk9491)					
1	Semester 1	Core	Part Time On Campus	CMM518	Security Testing		Muhammad Shadi Hajar	Yes	1.0	15	Level 11	
1	Semester 1	Core	Full Time On Campus	CMM518	Security Testing		Muhammad Shadi Hajar	Yes	1.0	15	Level 11	
1	Semester 2	Core	Part Time On Campus	CMM523	Database and Web Security		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11	
1	Semester 2	Core	Full Time On Campus	CMM523	Database and Web Security		Hatem Ahriz (cmsha)	Yes	2.0	15	Level 11	
1	Semester 2	Core	Full Time On Campus	CMM519	Incident Management and Forensics		Ian Harris (ish7850)	Yes	2.0	15	Level 11	
1	Semester 2	Core	Part Time On Campus	CMM528	Network Security		Christopher McDermott (cdm7692)	Yes	1.0	15	Level 11	
1	Semester 2	Core	Full Time On Campus	CMM528	Network Security		Christopher McDermott (cdm7692)	Yes	1.0	15	Level 11	
1	Semester 2	Core	Full Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11	
1	Semester 3	Core	Part Time On Campus	CMM542	Human Factors in Security		Christopher McDermott (cdm7692)	Yes	2.0	15	Level 11	
			Full									

1	Semester 3	Core	Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 3	Core	Part Time On Campus	CMM541	Machine Learning for Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11
1	Semester 4	Core	Part Time On Campus	CMM519	Incident Management and Forensics		Ian Harris (ish7850)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
1	Semester 5	Core	Part Time On Campus	CEM100	MSc Project	Routes A, B, C; undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 5	Elective	Part Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 5	Elective	Part Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Aberdeen, United Kingdom	On-Campus	Part Time	
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Sources of Further Information

No sources of further information recorded
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